

Eddy Current Formation and Its Applications in Conductive Plates

Simulation and Experimental Verification

1. Introduction

Eddy currents are circular electric currents induced in conductive materials when they are exposed to a changing magnetic field. These currents oppose the magnetic field that created them (Lenz's Law) and are used in many engineering applications such as non-destructive testing, electromagnetic braking, induction heating, and metal detection.

This report presents:

- The theory behind eddy current formation
- A Simulink-based lumped-model simulation
- Equivalent circuit behavior
- Real-life experimental verification procedure
- Observations and conclusions

2. Theory of Eddy Current Formation

When an AC-driven coil is placed near a conductive plate:

1. The alternating magnetic field from the coil passes through the plate.
2. According to Faraday's Law, a time-varying magnetic field induces EMF inside the plate.
3. Because the plate is conductive, this EMF generates circular (eddy) currents.
4. These currents oppose the original magnetic field (Lenz's Law).

- Eddy current strength depends on:
 - Conductivity (σ) of plate
 - Frequency (f) of excitation
 - Distance between coil and plate (lift-off)
 - Coil inductance and geometry

Skin depth:

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$$

- Shows how deeply current penetrates into the plate
- Higher frequency $\rightarrow$ smaller $\delta \rightarrow$ current flows near surface

3. Simulation Model (Lumped Eddy-Current Model)

To represent a conductive plate, a simplified two-loop coupled-inductor model is used.

3.1 Circuit Representation

- **Primary coil:** R_1 in series with L_1
- **Secondary (eddy current loop):** R_2 in series with L_2
- Coupling between coils represented by mutual inductance M

3.3 Parameter Values Used

Parameter	Value	Meaning
L_1	50 μH	Primary coil inductance
R_1	0.5 Ω	Copper coil DC resistance
L_2	2 μH	Eddy-current loop inductance
R_2	0.02 Ω	Effective resistance of plate loop

Coupling (k)	0.05	Represents lift-off
M	0.5 μ H	Mutual inductance
Source	1 V, sine	Drive voltage
Frequency	10 kHz (variable)	Excitation frequency

4. Simulation Results (Expected)

4.1 Primary Current (i_1)

- Sinusoidal waveform
- Phase slightly shifted due to inductive loading

4.2 Induced Eddy Current (i_2)

- Slightly delayed (lagging) sinusoidal
- Amplitude increases with frequency up to a point
- Too high frequency $\rightarrow$ inductive reactance increases $\rightarrow$ reduction in current

4.3 Power Dissipation in Plate (R_2)

- Represents real eddy current losses
- Increases with frequency until skin-depth limits current flow

4.4 Effect of Lift-Off (k value)

k Value	Observation
0.1	Strong eddy currents
0.05	Normal coupling
0.01	Eddy current small (coil far from plate)

5. Experimental Verification

5.1 Required Items

- Copper or aluminum plate
- Excitation coil (10–50 turns)
- Function generator (1–100 kHz)
- Oscilloscope or lock-in amplifier
- Optional: Pickup coil

5.2 Procedure

1. Connect primary coil to function generator.
2. Bring plate close to coil.
3. Observe primary current using a shunt resistor or current probe.
4. Move plate away and compare current change.
5. Place pickup coil on plate surface to measure induced voltage.
6. Sweep frequency and record values.

Expected Experimental Behavior

Condition	Observation
Plate near coil	Primary current drops slightly, induced voltage ↑
Plate far from coil	Induced voltage ↓, primary current ↑
Higher frequency	Stronger eddy currents → more heating
Very high frequency	Eddy current reduces due to skin effect

6. Applications of Eddy Currents

- **Non-destructive testing (NDT)** → crack detection
- **Induction heating** → cooking, metal hardening
- **Magnetic braking** → trains, amusement park rides
- **Metal sorting / detection** → recycling industries
- **Energy harvesting / sensors**

7. Conclusion

This study successfully demonstrates:

- Formation of eddy currents in conductive plates
- Their dependence on excitation frequency, resistance, inductance, and coupling
- Simulation using a lumped coupled-inductor model
- Real experimental observations matching theoretical predictions

Eddy currents are a fundamental and widely useful phenomenon, and the Simulink model provides an accessible method to visualize their behavior.

8. References

1. Faraday's Law of Electromagnetic Induction
2. Lenz's Law
3. MATLAB Simulink Documentation
4. Basic Electromagnetic Theory – Griffiths